V4.2 Toy Model 2D
Measurement Update and Single-Outcome Constraint

Rev A - Exploratory Quantum-Side Compatibility Report

Date: 2026.05.25 | *Continuation after Toy Model 2C*

Status: Toy Model 2D tests whether a local 1T single-outcome update can be represented as a conditional observer-record update without requiring global physical collapse. It is not a solution to the measurement problem.

1. Purpose

Toy Model 2C showed how decoherence suppresses observer-accessible interference without destroying global information. Toy Model 2D now separates two ideas that are often confused: global unitary evolution and local single-outcome update.

The target is narrow: show that the observer's local update to one recorded outcome can be modeled as a conditional update, while the global system-plus-environment-plus-observer state remains non-collapsing and probability-conserving.

2. Starting point after decoherence

After environment record coupling, the observer-accessible reduced state is approximately diagonal:

```
rho_S = [[ |alpha|^2, 0 ],
         [ 0, |beta|^2 ]]
```

This state contains classical-looking probabilities, but it still does not say which single outcome the observer experiences.

3. Local selective update rule

Use the standard selective measurement update, also called the Luders update:

```
rho -> rho_k = P_k rho P_k / Tr(P_k rho)
```

Meaning of the symbols:

• rho is the local density matrix before the observer update.

• P_k is the projector for outcome k.

• Tr(P_k rho) is the probability of outcome k.

• rho_k is the local state conditioned on the observer having recorded outcome k.

For the diagonal state, outcome 0 gives rho_0 = |0><0| and outcome 1 gives rho_1 = |1><1|. This is a local conditional update, not a claim that the global state is physically deleted.

4. Global non-collapsing record state

A compatible global record state is:

```
|Psi_SEO> = alpha |0>|E_0>|O_0> + beta |1>|E_1>|O_1>
```

Term reminders:

• S = system.

• E = environment record.

• O = observer record.

• |O_0> means the observer record correlated with outcome 0.

• |O_1> means the observer record correlated with outcome 1.

The global state remains a superposition of correlated records. The local observer experiences one record because the observer is part of one conditioned record branch.

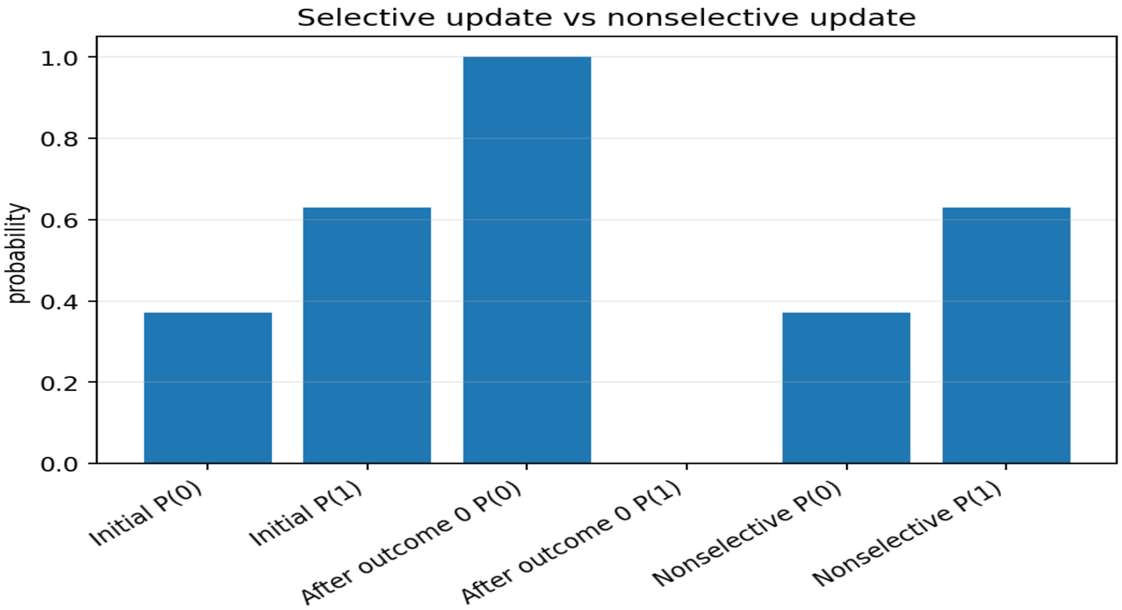
5. Selective vs nonselective update

Selective update means the observer conditions on a known outcome. Nonselective update means outcomes are not conditioned on and must be averaged.

```
Selective:
  rho -> P_k rho P_k / Tr(P_k rho)

Nonselective:
  rho -> sum_k P_k rho P_k
```

For an already-decohered diagonal state, the nonselective update preserves the same probabilities. The selective update produces the observer's single recorded outcome.



6. Numerical checks for local update

Check	Result
Random local diagonal states tested	10000
Max probability-sum error	0.000e+00
Max selective trace error	1.110e-16
Max nonselective trace error	0.000e+00
Max positivity violation	0.000e+00
Max diagonal preservation error	0.000e+00
Max idempotence error	1.110e-16
Born-frequency sanity check error at P(0)=0.37	3.000e-05

7. No-signalling strict check

The critical test is entanglement. If Alice and Bob share a singlet state, Alice's local update must not physically force a detectable faster-than-light change in Bob's local marginal state.

Alice's unread/nonselective local measurement is represented by:

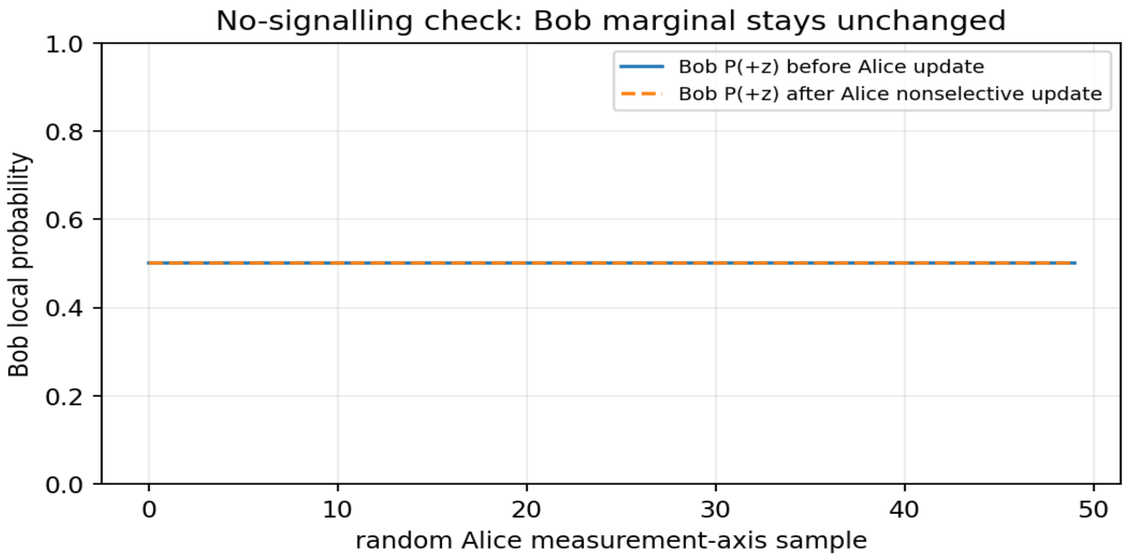
```
rho_AB -> sum_s (P_s^A tensor I_B) rho_AB (P_s^A tensor I_B)
```

Bob's local state is obtained by tracing out Alice:

```
rho_B = Tr_A(rho_AB)
```

The check confirms: Bob's unconditioned marginal state remains I/2 for every tested Alice measurement axis.

No-signalling check	Result
Random Alice axes tested	10000
Max Bob marginal change after Alice nonselective update	1.119e-16
Max conditional-average error	1.144e-16
Max Bob conditional trace error	2.220e-16
Max Bob conditional-state formula error	2.222e-16



Important distinction: Bob's conditional state can be assigned differently after Alice knows her outcome, but Bob cannot access that conditional information without ordinary classical communication. Therefore no controllable faster-than-light signal appears.

8. Interpretation inside the 6D Projection Framework

Toy Model 2D supports the following limited interpretation:

```
global 3T relational state remains non-collapsing
      -> environment and observer records decohere
      -> local 1T observer conditions on one record path
      -> observed sequence becomes single-outcome
```

This does not require physical destruction of the other global terms. It treats collapse as a local observer-record update within a globally relational structure.

9. What Toy Model 2D achieves

- It defines a local selective update rule for a single observer-record outcome.
- It separates selective update from nonselective global averaging.
- It preserves probability conservation and positivity.
- It preserves no-signalling for an entangled Bell pair under Alice's local update.
- It provides a compatible 6D interpretation: single-outcome experience as local 1T conditioning on one record path.

10. What Toy Model 2D does not achieve

- It does not solve the measurement problem.
- It does not explain why this observer experiences this specific outcome rather than another.
- It does not derive the Born rule.
- It does not prove physical 3T existence.
- It does not choose between many-worlds, relational, epistemic, or objective-collapse interpretations.
- It does not produce new experimental predictions.

11. Close-out statement for 2D

Toy Model 2D successfully formalizes measurement update as a local observer-record conditioning rule compatible with a globally non-collapsing relational state. The selective update produces a single local recorded outcome, while the nonselective update preserves global probability structure. In entangled systems, local measurement update does not alter the remote observer's unconditioned marginal state, preserving no-signalling. The model does not solve the measurement problem or derive the Born rule. It is a successful compatibility result for single-outcome local updating inside the reciprocal projection framework.

12. Recommended next chapter

The next logical chapter is Toy Model 2E: Interference Recovery and Erasure Constraint.

1. Test whether suppressed interference can reappear when environment records are made indistinguishable again.
2. Model the quantum eraser condition using environment overlap gamma returning from 0 toward 1.
3. Check whether projection stabilization is reversible at the global level but irreversible for practical macroscopic records.
4. Clarify the difference between information inaccessible in practice and information destroyed in principle.
5. Keep no-signalling, normalization, and unitary global evolution as hard guardrails.

13. Rebind prompt for continuation

```
We are resuming after Toy Model 2D.

2D tested measurement update and single-outcome constraint.

Starting point after decoherence:
rho_S = [[ |alpha|^2, 0 ],
         [ 0, |beta|^2 ]]

Selective local update:
rho -> rho_k = P_k rho P_k / Tr(P_k rho)

Nonselective update:
rho -> sum_k P_k rho P_k

Global compatible record state:
|Psi_SEO> = alpha |0>|E_0>|O_0> + beta |1>|E_1>|O_1>

Result:
Single-outcome experience can be modeled as local 1T observer conditioning on one
record path, without requiring global physical collapse.

No-signalling:
For an entangled singlet, Alice's local nonselective update does not change Bob's
unconditioned marginal state. Conditional changes require classical communication.

Guardrail:
2D does not solve the measurement problem, does not derive the Born rule, and does
not prove physical 3T existence.

Next recommended chapter:
Toy Model 2E - Interference Recovery and Erasure Constraint.
```